3D Printing and Additive Manufacturing

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Introduction to 3D Printing

3D printing, technically called additive manufacturing, is a process of creating a three-dimensional physical object from a digital design by adding material layer upon layer until the object is complete. The word "additive" distinguishes it from traditional subtractive manufacturing, such as milling, drilling and turning, in which material is cut away from a solid block. Additive manufacturing is also different from casting and moulding, which need a mould. Because no special tooling is needed, 3D printing makes it economical to produce single prototypes, customised products and complex shapes that are impossible with conventional methods, such as internal channels and lattice structures.

History

Charles "Chuck" Hull invented stereolithography, the first 3D printing technology, in 1983 and patented it in 1986; he then founded the company 3D Systems. In 1989 Scott Crump invented Fused Deposition Modeling (FDM) and founded Stratasys. Selective Laser Sintering (SLS) was developed at the University of Texas in the 1980s. The early 2000s saw 3D printing in medicine and, after key FDM patents expired around 2009, the arrival of low-cost desktop printers such as the RepRap project and MakerBot, which made 3D printing accessible to students and hobbyists.

How 3D Printing Works

The workflow has several steps. First, a 3D model is created using Computer-Aided Design (CAD) software such as Fusion 360, SolidWorks or Tinkercad, or captured with a 3D scanner. The model is exported in a file format such as STL or 3MF. Next, slicing software, such as Cura or PrusaSlicer, cuts the model into hundreds or thousands of thin horizontal layers and generates G-code, the machine instructions. The printer then builds the object layer by layer. Finally, post-processing removes support structures and may include sanding, curing, painting or heat treatment.

Main Types of 3D Printing Technologies

Fused Deposition Modeling (FDM), also called Fused Filament Fabrication, melts a plastic filament and extrudes it through a heated nozzle; it is the most common and cheapest method. Stereolithography (SLA) and Digital Light Processing (DLP) use light to cure liquid photopolymer resin layer by layer and give very smooth, detailed parts. Selective Laser Sintering (SLS) uses a laser to fuse powder, typically nylon, without needing supports. Metal technologies such as Direct Metal Laser Sintering (DMLS) and Selective Laser Melting (SLM) melt metal powder with a laser to make strong metal parts for aerospace and medicine. Binder jetting and construction 3D printing, which extrudes concrete, are other forms.

3D Printing Materials

Two common 3D printing materials are PLA (polylactic acid) and ABS (acrylonitrile butadiene styrene). PLA is a biodegradable plastic made from plant starch such as corn, easy to print and popular among beginners. ABS is a stronger, heat-resistant plastic used for durable parts, similar to the material of LEGO bricks. Other materials include PETG, nylon, flexible TPU, photopolymer resins, ceramics, carbon-fibre composites, metal powders such as titanium, stainless steel, aluminium and Inconel, and even concrete, chocolate and biological cells.

Medical Uses of 3D Printing

Medicine is one of the most important application areas. One medical use of 3D printing is making custom prosthetics, such as artificial limbs and hands, shaped exactly to a patient's body at low cost. It is also used for patient-specific titanium implants for hips, skulls and jaws, dental crowns and clear aligners, surgical guides, hearing aids and anatomical models printed from CT or MRI scans that help surgeons plan complex operations. Researchers are working on bioprinting, which prints living cells and tissues, with the long-term hope of printing organs.

Other Applications

Aerospace companies print lightweight fuel nozzles and rocket engine parts. Automotive firms print prototypes and spare parts. Architects print scale models and even houses. Consumers print toys, tools, phone cases and replacement parts. In education, schools use 3D printers to teach design and engineering.

3D Printing in India

Chennai-based startup Agnikul Cosmos, incubated at IIT Madras, launched the world's first rocket with a fully 3D-printed single-piece semi-cryogenic engine, Agnibaan SOrTeD, from Sriharikota in May 2024. IIT Madras-incubated Tvasta built a 3D-printed house in Chennai in 2021. In 2023 Larsen & Toubro built India's first 3D-printed post office in Bengaluru. The Government's National Strategy on Additive Manufacturing aims to increase India's role in the global market. Many Indian engineering colleges and makerspaces have FDM printers.

Challenges and Future

Challenges include slower speed for mass production, limited size, material cost, surface finish, strength differences between layers and the need for skilled designers. The future includes faster metal printing, multi-material printing, 4D printing, in which objects change shape over time, on-demand spare parts and sustainable recycled materials.

Quick Facts

- Additive manufacturing builds objects by adding material layer by layer from a digital model.
- PLA and ABS are two common 3D printing materials.
- 3D printing is used in medicine for prosthetics, implants, dental aligners and surgical models.
- Chuck Hull invented stereolithography; FDM was invented by Scott Crump in 1989.
- Agnikul Cosmos of Chennai launched a rocket with a 3D-printed engine in May 2024.